

Software Requirements Specification (SRS)

Project Title

ShopFlow (*Working Title*)

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for an e-commerce platform designed for managing and selling products online. The system aims to provide customers with a seamless shopping experience while enabling administrators to efficiently manage inventory, orders, customers, and business operations.

1.2 Objectives

- Build a modern, responsive e-commerce platform.
 - Provide a secure online shopping experience.
 - Enable efficient inventory and order management.
 - Design the system for scalability and maintainability.
 - Support future expansion to mobile apps and additional sales channels.
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2. Scope

The application will provide:

Customer Portal

- Browse products
- Search and filter products
- View product details
- Add items to cart
- Checkout
- Secure online payment
- Order tracking
- User profile management

Administration Portal

- Dashboard
 - Product management
 - Category management
 - Inventory management
 - Order management
 - Customer management
 - Coupon management
 - Sales reporting
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3. User Roles

Guest

- Browse products
- Search products
- Register
- Login

Customer

- Manage profile
- Place orders
- View order history
- Track orders
- Save addresses

Administrator

- Manage products
 - Manage inventory
 - Process orders
 - Issue refunds
 - Manage customers
 - View analytics
 - Configure settings
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4. Functional Requirements

Authentication

- User registration
 - Secure login
 - Password reset
 - Email verification (optional)
 - JWT authentication
 - Role-based authorization
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Product Management

Administrator can:

- Create products
 - Edit products
 - Delete products
 - Upload images
 - Assign categories
 - Set pricing
 - Manage stock
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Product Catalog

Customers can:

- Browse categories
 - Search products
 - Filter by category
 - Filter by price
 - Sort products
 - View product details
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Shopping Cart

- Add items
- Update quantity
- Remove items
- Save cart
- Calculate subtotal

- Calculate taxes
 - Calculate shipping
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Checkout

- Select shipping address
 - Select payment method
 - Apply coupon
 - Review order
 - Confirm purchase
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Payment

Support:

- UPI
- Credit/Debit Cards
- Net Banking
- Wallets

System should:

- Verify payment
 - Record transactions
 - Handle failed payments
 - Process refunds
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Orders

Customers can:

- View order history
- Track shipment
- Download invoices

Administrators can:

- View orders
 - Update status
 - Cancel orders
 - Refund payments
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Inventory

System should:

- Track stock
 - Reserve stock during checkout
 - Update stock after purchase
 - Notify low inventory
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Notifications

Email:

- Order confirmation
- Payment confirmation
- Shipping notification
- Password reset

Future:

- SMS
 - WhatsApp
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5. Non-Functional Requirements

Performance

- Page load < 2 seconds
 - API response < 300 ms (typical)
 - Image optimization
 - CDN support
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Scalability

Support:

- Multiple warehouses
 - Thousands of products
 - High concurrent users
 - Horizontal scaling
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Security

- HTTPS
 - Password hashing
 - Rate limiting
 - Input validation
 - SQL injection prevention
 - XSS protection
 - CSRF protection (where applicable)
 - Audit logging
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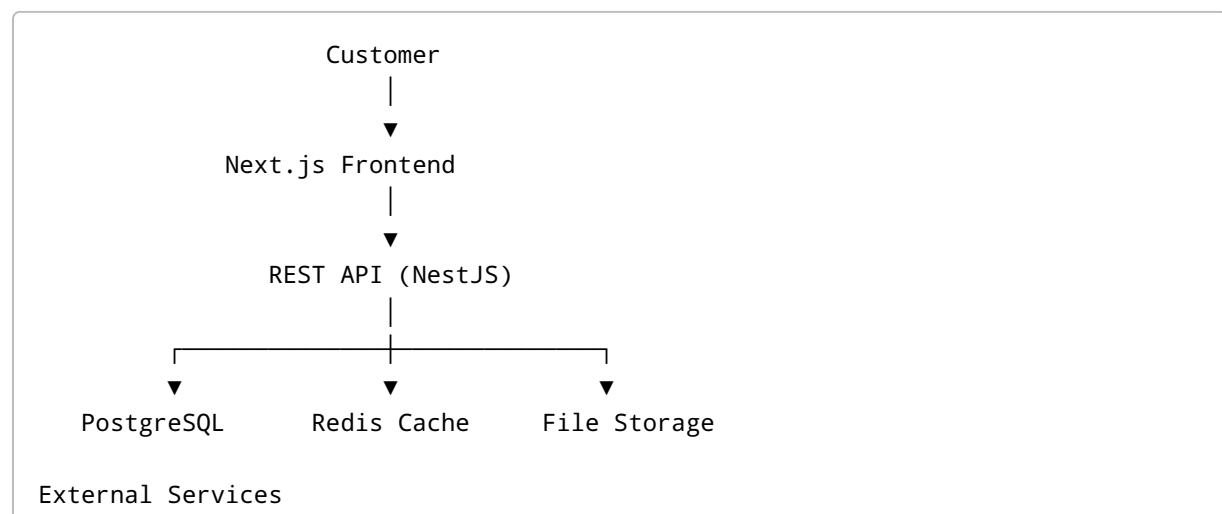
Reliability

- Daily backups
 - Error monitoring
 - Graceful error handling
 - Automatic recovery where feasible
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Maintainability

- Modular architecture
 - Clean code
 - Unit tests
 - Integration tests
 - API documentation
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6. System Architecture



Payment Gateway
Email Service
SMS Service
Shipping Provider
Analytics

7. Database Entities

Core tables:

- Users
- Roles
- Products
- Categories
- ProductImages
- Inventory
- Orders
- OrderItems
- Addresses
- Payments
- Coupons

Future tables:

- Reviews
- Wishlist
- LoyaltyPoints
- GiftCards
- Referrals

8. API Modules

Authentication

- POST /auth/register
- POST /auth/login
- POST /auth/refresh
- POST /auth/logout

Products

- GET /products

- GET /products/:id
- POST /products
- PATCH /products/:id
- DELETE /products/:id

Categories

- CRUD endpoints

Cart

- GET /cart
- POST /cart/items
- PATCH /cart/items/:id
- DELETE /cart/items/:id

Orders

- POST /orders
- GET /orders
- GET /orders/:id

Payments

- POST /payments/create
- POST /payments/webhook

Admin

- Dashboard
- Reports
- Inventory
- Users

9. Suggested Project Structure

```
apps/  
  web/  
  api/  
  
packages/  
  ui/  
  config/  
  types/  
  shared/
```



```
services/  
  payment/  
  email/  
  notification/  
  
database/  
  prisma/  
  
docs/  
  
tests/
```

10. Development Roadmap

Milestone 1

- Project setup
- Authentication
- Database
- User roles

Milestone 2

- Product management
- Categories
- Inventory

Milestone 3

- Product catalog
- Search
- Product pages

Milestone 4

- Shopping cart
- Checkout
- Payment integration

Milestone 5

- Orders
- Customer profile
- Email notifications

Milestone 6

- Admin dashboard
- Reporting
- Coupons

Milestone 7

- Deployment
 - Monitoring
 - SEO
 - Analytics
 - Performance tuning
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11. Future Enhancements

- Wishlist
 - Product reviews
 - Loyalty program
 - Gift cards
 - Referral system
 - AI product recommendations
 - Multi-vendor marketplace
 - Multi-language support
 - Multi-currency support
 - Native mobile applications
 - POS integration
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12. Success Criteria

The project will be considered successful when it:

- Provides a smooth shopping experience.
- Handles payments securely.
- Enables administrators to manage products and orders efficiently.
- Is scalable for future business growth.
- Can support additional platforms with minimal architectural changes.